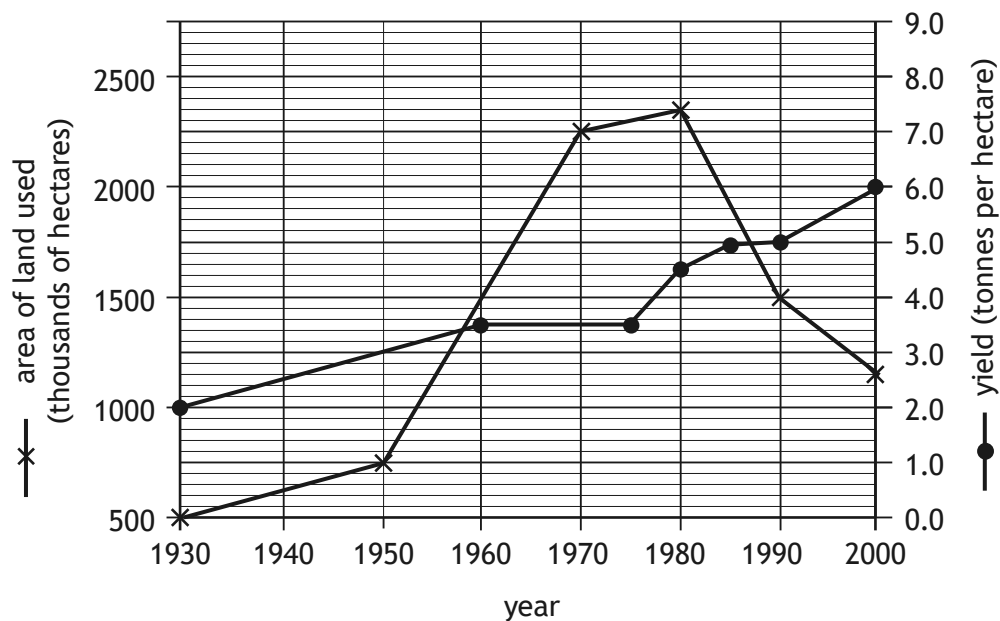


13. The graph shows the area of land in the UK used to grow barley, and its yield between 1930 and 2000.



- (a) Use values from the graph to describe the changes in the area of land used to grow barley from 1950 to 2000.

2

- (b) State the barley yield when the area of land used was 750 thousand hectares.

1

_____ tonnes per hectare

[Turn over



13. (continued)

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- (c) Barley is used to feed livestock.

The table shows the mass of different types of livestock produced in the UK between 2000 and 2020.

Year	Mass of livestock produced in the UK ($\times 10^6$ kg)		
	Cattle	Pigs	Poultry
2000	960	1293	1704
2010	1076	1354	1770
2020	1104	1380	2348

Calculate the percentage increase in the mass of cattle produced between 2000 and 2020.

1

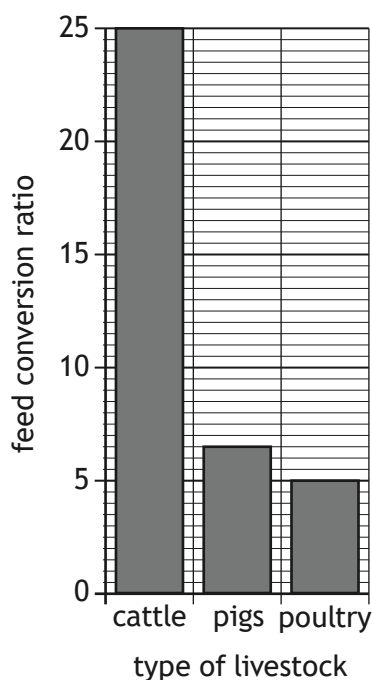
Space for calculation

_____ %

- (d) The feed conversion ratio for different types of livestock can be calculated using the following formula.

$$\text{Feed conversion ratio} = \frac{\text{mass of food eaten by livestock}}{\text{mass of livestock produced}}$$

The bar chart shows the feed conversion ratio of different types of livestock.



13. (d) (continued)

Use information in the table and bar chart to calculate the total mass of feed required for poultry production in the UK in 2020.

1

Space for calculation

_____ $\times 10^6$ kg

(e) Using information in the bar chart explain why eating poultry or pigs reduces the impact on food security compared with eating cattle.

2

[Turn over

